SYNCLAB

Centralized Workshop & Laboratory Allocation Platform

Group 01
NANO 2142 Introduction to Software Development

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Scheduling problem

Students and laboratory administrators need a shared view of available sessions.

Availability

A student needs a laboratory, a suitable date and one or more free time slots.

Approval

The administrator must review each request and record a decision.

Record keeping

Booking details and status must remain available for later changes.

Objectives and implemented features

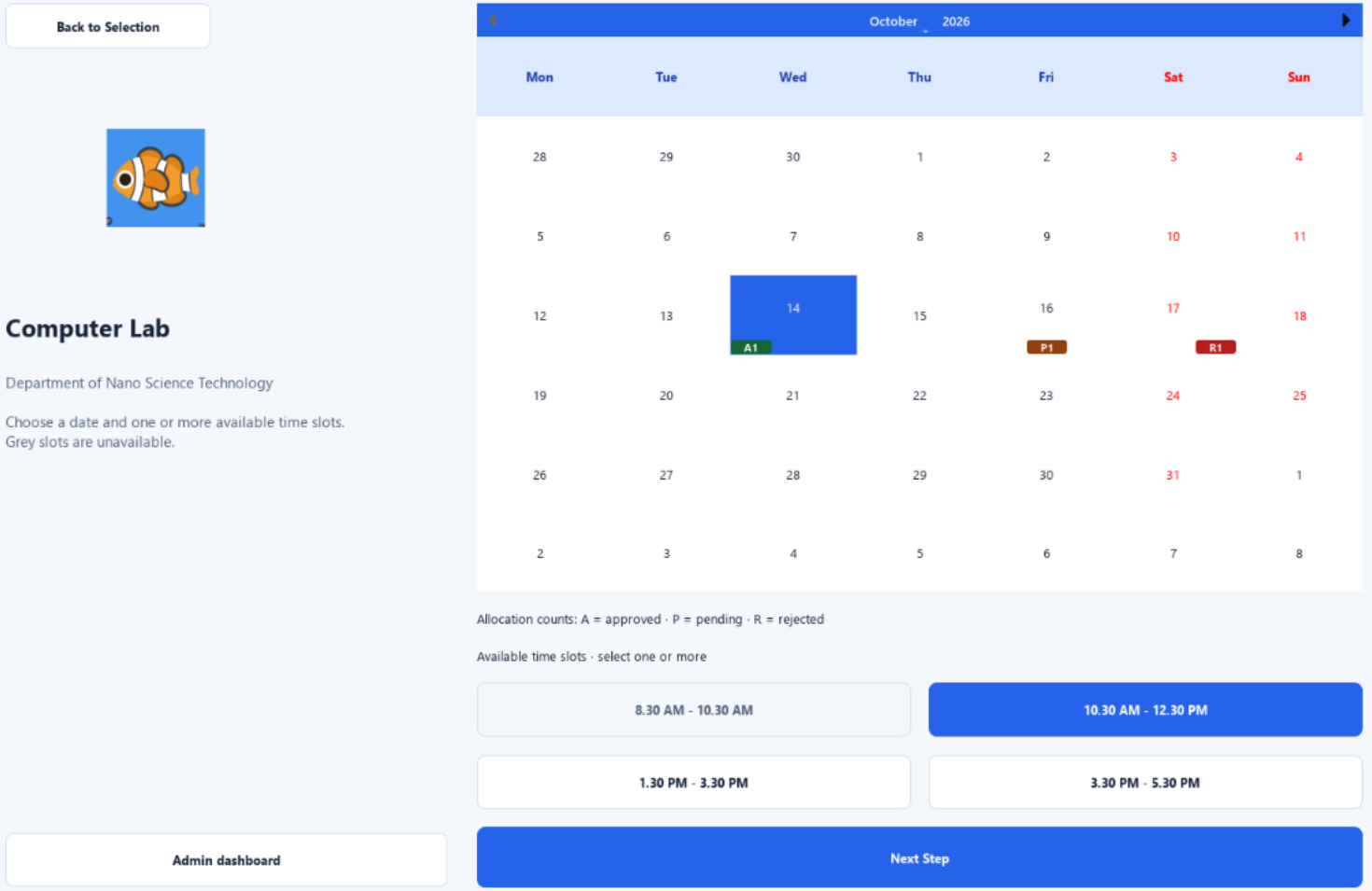
Provide one desktop workflow for laboratory allocation and administrator review.

Scope	Implemented behavior
10 laboratories and workshops	One category per allocation across four departments
4 daily time slots	Students can select more than one available slot
Student self service	Create, update and cancel with an allocation key
Administrator review	Date filter, remarks, status decisions and record editing
Local persistence	SQLite stores allocations and normalized related facts

Application screens

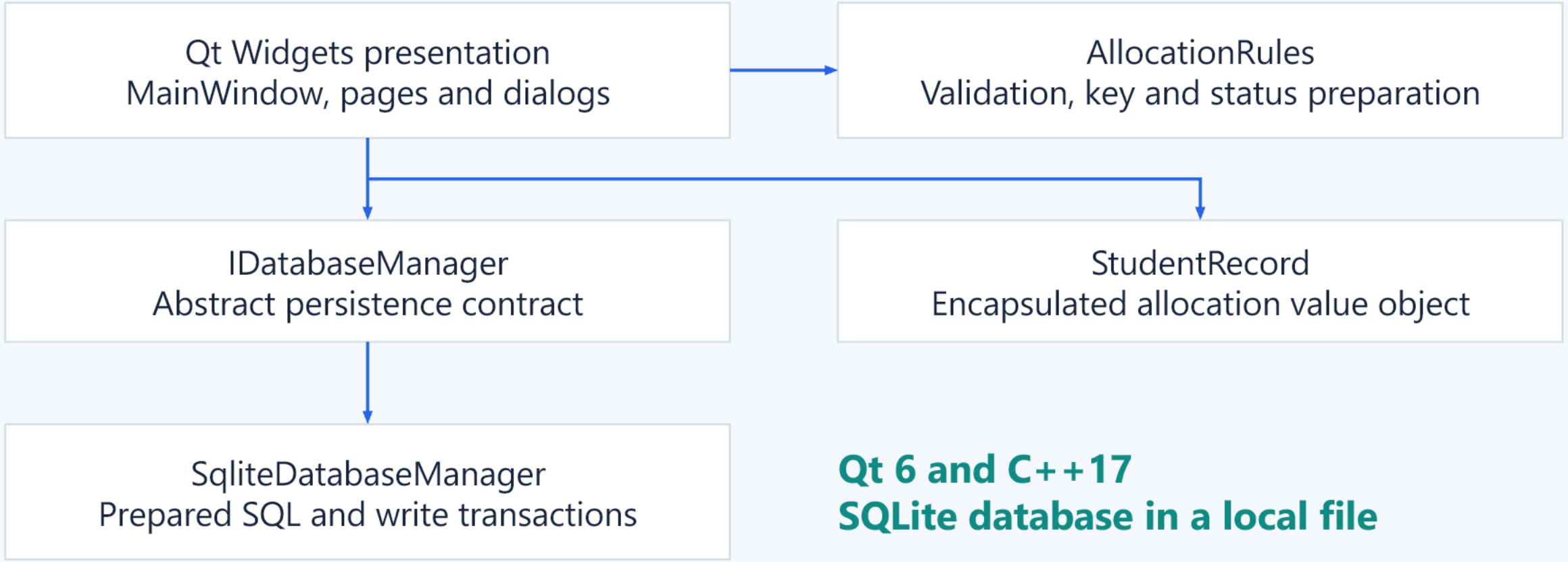
Welcome
Workshop selection
Date and time
Allocation form
Admin dashboard

The five pages share the same theme and navigation state.



Actual calendar view with fictional demonstration data

Application architecture



Qt 6 and C++17
SQLite database in a local file

Database design

Relation	Key	Stored facts
students	index_no	Name, student department and email
workshops	category	Laboratory ownership department
records	id	Student/workshop references, project, supervisor, date and status
booking_slots	record_id + time_slot	One row for each selected slot

One student and one workshop can each have many allocations.
Deleting an allocation also deletes its slot rows.

Separate profile, workshop and slot facts reduce repeated data.

Object oriented design

Encapsulation

StudentRecord exposes private data through getters and setters.

Abstraction

MainWindow calls IDatabaseManager without issuing SQL directly.

Inheritance

SqliteDatabaseManager implements the database interface. Qt widgets extend base classes.

Polymorphism

Virtual database methods select the SQLite implementation at runtime.

Event design

Qt signals and slots connect user actions to workflow methods.

Allocation workflow

1. Choose laboratory
2. Select date and free slots
3. Enter student and project details
4. Validate and save the allocation
5. Administrator reviews the request

Pending

A new allocation reserves its selected slots while the administrator reviews it.

Approved or Rejected

Approval keeps the slots reserved. Rejection makes them available again.

A student update returns the request to Pending.

Testing and observed results

10 of 10 test groups passed

69 assertions, 0 failed assertions. Qt 6.11.2 on offscreen.

Evidence scope	Observed checks
Allocation rules	Required inputs, dates, key formatting and edit status
SQLite operations	CRUD, conflict rejection, status transitions and filters
Live Qt workflow	Navigation, invalid email, successful save and admin access
Data safety	Isolated test databases and fictional demonstration records

Email delivery, stress testing and field deployment still need separate validation.

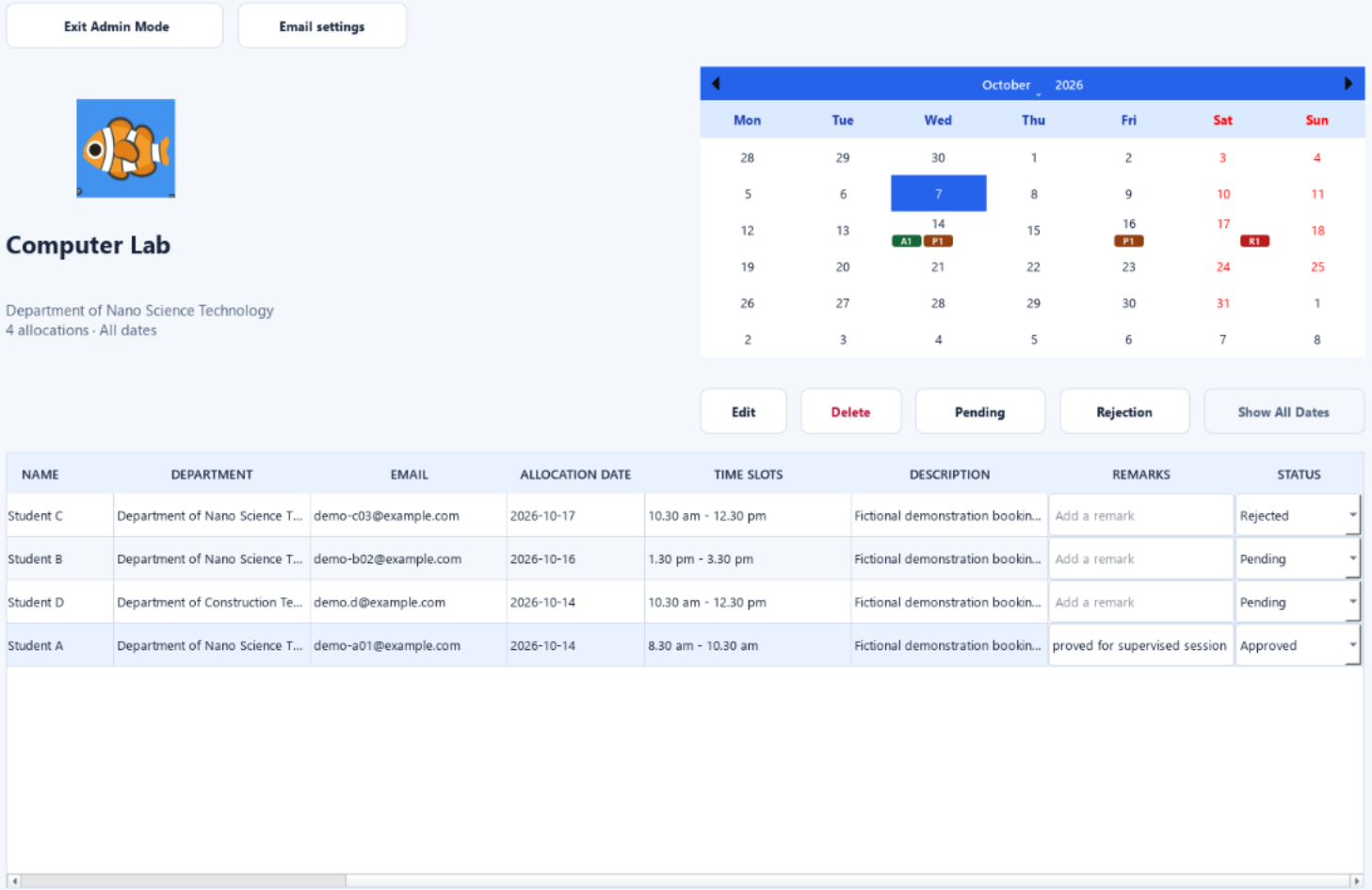
Limitations and future work

Current limitation	Proposed improvement
Local desktop database	Server API and shared multi-device access
Four-digit allocation keys	Larger keys and stronger student identity checks
Shared administrator passcodes	Named accounts, hashed credentials and access logs
External email drafts	Controlled notification service and delivery records
Fixed catalog and slot schedule	Administrator-managed schedules and closures
Manual operational backup	Guided backup, restore and allocation report export

Demonstration sequence

1. Select Computer Lab
2. Show a reserved slot
3. Submit a valid allocation
4. Open administrator view
5. Review status and date filter

Use fictional records for the assessment demonstration.



Actual administrator dashboard with demonstration records

Team presentation and conclusion

Member names and signed contribution declaration are pending.

Proposed speaking responsibilities

Project scope and demonstration
Qt interface and allocation rules
Database design and object oriented implementation
Testing, limitations and future development

SYNCLAB combines allocation requests, availability checks and administrator decisions in one desktop workflow.